BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Ethical, Cultural and Environmental Impacts

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does a compression utility do?

2. What does a denial-of-service attack try to do?

A) Make a service unavailable
B) Improve network speed
C) Create secure backups
D) Repair damaged hardware

3. "Data interception is only copying a paper document from a filing cabinet." Correct this. Explain why.

4. A new user needs to open and sort files. "A user interface replaces all CPU instructions." Correct this. How does an interface help?

Today's objectives

Explain the core ideas of ethical, cultural and environmental impacts using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Ethical impacts

Consider fairness, privacy and responsible uses of collected data.

Cultural impacts

Digital technology changes communication, access to information and how groups participate.

Environmental impacts

Device manufacture, energy usage and electronic waste all have costs.

Stakeholders

A balanced evaluation considers benefits, harms and who experiences each effect.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Ethical impacts	Consider fairness, privacy and responsible uses of collected data.
Cultural impacts	Digital technology changes communication, access to information and how groups participate.
Environmental impacts	Device manufacture, energy usage and electronic waste all have costs.
Stakeholders	A balanced evaluation considers benefits, harms and who experiences each effect.

I Do - One worked example

Evaluate a school app that tracks pupil location all day. Give a balanced response.

1. Benefit: can support safety or locating missing equipment if appropriate.
2. Risk: constant tracking may intrude on privacy.
3. Identify need for proportionality, consent/legal basis and secure access.
4. Give a justified judgement about limited, transparent use.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Discuss replacing paper letters with a school app.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write the missing ELSE output for a user without internet access.

```
hasInternet = false
if hasInternet then
print("Use website")
else
...
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Explain one environmental disadvantage of frequent device replacement. [3 marks]

Q3. Give an ethical concern about large-scale personal-data collection. [3 marks]

Q4. Explain one benefit and one disadvantage of online-only school communication. [4 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A balanced evaluation considers benefits and harms for stakeholders. ☐ T ☐ F
2. Consider fairness, privacy and responsible uses of collected data. ☐ T ☐ F
3. Digital technology changes communication, access to information and how groups participate. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 It reduces the storage space required for data, using a suitable lossless or lossy method. ☐

Q2 A - Make a service unavailable ☐

Q3 Capturing data during transmission, potentially exposing confidential information. ☐

Q4 A user interface lets people use the computer, through GUI controls or typed commands. Other OS tasks manage devices, accounts, memory and files. ☐

Code First - Model

```
hasInternet = false
if hasInternet then
print("Use website")
else
print("Offer telephone access")
endif
```

Output: Offer telephone access

We Do - Check the model

1. Provide a balanced conclusion. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Creates electronic waste and uses energy/resources in manufacture. ☐

Q3 Privacy, informed use/consent or biased profiling, with contextual explanation. ☐

Q4 Faster cheaper access for connected households; excludes families without dependable devices/internet. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: